

Table C3.8.1.2.1-2—Wind Pressure on the Bridge Structures for Strength V and Service I Load Combinations

Load Combination	Wind Pressure on the Structure, P_Z , for the Specified Wind Speed (ksf)
Strength V	$0.0163C_D$
Service I	$0.0125C_D$

The pressure exposure and elevation coefficient, K_Z , for Strength III and Service IV load combinations shall be determined as follows:

$$K_Z(B) = \frac{\left[2.5 \ln \left(\frac{Z}{0.9834} \right) + 6.87 \right]^2}{345.6} \quad (3.8.1.2.1-2)$$

$$K_Z(C) = \frac{\left[2.5 \ln \left(\frac{Z}{0.0984} \right) + 7.35 \right]^2}{478.4} \quad (3.8.1.2.1-3)$$

$$K_Z(D) = \frac{\left[2.5 \ln \left(\frac{Z}{0.0164} \right) + 7.65 \right]^2}{616.1} \quad (3.8.1.2.1-4)$$

where:

$K_Z(B)$, $K_Z(C)$, and $K_Z(D)$ are K_Z for wind exposure Category B, C, and D, respectively.

The structure height, Z , used in determining the pressure exposure and elevation coefficient, K_Z , shall be taken as:

- For bridge superstructures: the average height of the top of the superstructure above the surrounding ground or water surface.
- For bridge substructures not extending above the elevation of the superstructure: unless otherwise approved by the Owner, the height used in determining the wind pressure on the superstructure.
- For bridge substructures extending above the elevation of the superstructure: unless otherwise approved by the Owner, the height of the top of the substructure.

In the case of a long multi-span bridge with large variation in the ground surface elevation under the bridge, such as a bridge crossing a valley, the structure height, Z , may be varied from a span to span. For each span, the structure height, Z , may be taken as the largest value in the span.

Determining the wind pressure on substructures not extending above the elevation of the superstructure using the structure height used to determine the wind pressure on the superstructure results in slightly conservative values for most substructures. For extremely tall substructures, using a different height, including varying the height used for different segments of the substructure, may be allowed with the approval of the Owner.

Substructures extending above the elevation of the superstructure are typically associated with cable-stayed bridges and suspension bridges. Wind loads on such structures are typically determined using a structure-specific wind tunnel test.